Shunt Valve Artifact Detection and Catheter Length Estimation in MRI

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Abstract

This report presents a hybrid framework for automated catheter shunt analysis in 3D MRI, combining classical geometric processing with deep learning. A geometric prior served as a spatial seed for a 3D Residual U-Net, improving artifact segmentation Dice from 0.373 to 0.611. Catheter segmentation achieved a Dice of 0.8165 using a 3D U-Net, and a graph-based length estimation pipeline with a learned hidden length regressor yielded a mean absolute error of 1.73 mm.

1 Introduction

A ventriculoperitoneal (VP) shunt drains excess cerebrospinal fluid (CSF) from the brain's ventricles to the abdominal cavity via a thin catheter and metallic valve assembly (Prince, 2026). While lifesaving, the metallic valve produces MRI susceptibility artifacts that obscure surrounding structures, and the catheter's precise trajectory through brain parenchyma is critical for effective drainage and minimal tissue damage (Prince, 2026).

This work addresses three interrelated tasks defined in (Prince, 2026): valve artifact segmentation, catheter segmentation, and catheter length estimation, using a hybrid framework combining classical geometric processing with deep learning.

2 Background and Motivation

Metallic implant artifacts in MRI can obscure surrounding anatomy and reduce diagnostic accuracy, motivating automated detection and segmentation methods. Earlier approaches relied on thresholding, filtering, and morphological operations, often requiring substantial manual tuning and limited robustness across acquisition settings (Soille, 2003). More recently, convolutional neural networks (CNNs) have improved artifact segmentation by learning discriminative representations directly from annotated volumes, reducing dependence on handcrafted features (Litjens, 2017).

A major challenge in segmenting metallic artifacts and catheters is severe class imbalance, as these structures occupy only a small fraction of the image volume. To address this, prior work has adopted overlap-based losses such as Dice loss (Milletari et al., 2016), hard-example weighting through focal loss (Lin, 2017), and topology-preserving objectives such as clDice, which are particularly effective for thin tubular structures (Shit, 2021).

Catheter segmentation within brain MRI presents additional difficulties due to low contrast, thin geometry, and artifact-induced signal distortions. Early methods relied on geometric priors and centerline tracking, whereas modern deep learning approaches based on 3D U-Net architectures have enabled more reliable automatic delineation in MRI-guided interventions (Çiçek, 2016; Isensee, 2021). In particular, (Zaffino, 2020) demonstrated automated catheter identification in brain MRI using deep learning, highlighting the feasibility of robust catheter localization despite imaging artifacts and anatomical variability.

3 Data

This study utilized a dataset of 200 MRI volumes accompanied by three primary ground-truth components for each subject: a binary artifact mask delineating the signal void caused by the valve, a binary catheter mask tracking the 3D trajectory of the shunt, and a scalar length measurement (in mm) representing the true physical extent of the hardware. The cohort reflects clinical diversity, containing 190 right-sided and 10 left-sided shunt placements, with varying head orientations and intensity profiles across the scans.

4 Methodology

The proposed methodology implements a fully automated, end-to-end pipeline that sequentially performs valve artifact segmentation, catheter segmentation, and catheter length estimation to provide a complete quantitative analysis of the implanted hardware.

4.1 Preprocessing

The preprocessing pipeline was designed to standardize intensities and spatial orientation while preserving the integrity of the hardware. We applied Min-Max normalization, unsharp masking for contrast enhancement, and N4 bias field correction via ANTs

For spatial alignment to the MNI template, nonlinear SyN registration was initially explored but caused geometric distortion of the artifact region. We therefore adopted rigid-body registration with Lanczos windowed-sinc interpolation, which standardised coordinates while preserving hardware detail. Skull-stripping was excluded entirely, as BET (Smith, 2002) and SynthStrip (Hoopes et al., 2022) removed approximately 80% of the valve artifact in preliminary experiments.

4.2 Shunt Valve Artifact Segmentation

Initially, we aim to leverage the distinct geometric properties of the valve artifact which is characterized by a signal void "blob" at the brain boundary and is most prominent in axial views. The process begins by identifying an **axial anchor slice** representing a fully formed brain section while avoiding the irregular top of the skull or neck. This is enforced by selecting a slice with a **solidity** $S > 0.98$, defined as:

$$S = \frac{\text{Area}}{\text{Convex Area}} \tag{1}$$

This high threshold ensures the brain cross-section is a stable, near-perfect convex ellipse, allowing the algorithm to reliably detect the hardware-induced "dent" in the boundary. Once identified, standard contour detection is applied to the brain boundary, and a **convex hull** is calculated to bridge the indentation, creating a synthetic "complete" brain boundary.

By performing a **geometric subtraction** between the convex hull (H) and the original contour (C), the specific artifact region (A) is isolated:

$$A = H \setminus C \tag{2}$$

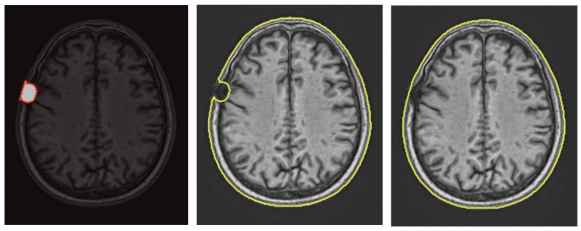


Figure 1: Geometric artifact isolation process. Left: The isolated binary artifact region (A). Middle: Original brain contour (C). Right: Computed convex hull (H).

Given that the shunt valve is a continuous 3D structure, the pipeline then propagates this region across the axial axis using a **three-slice sliding window** for spatial context, continuing until the void consistently diminishes. These axial segments serve as seeds for tracking in the sagittal and coronal planes, followed by **morphological smoothing** to remove jagged ridges and ensure a refined, volumetric binary mask.

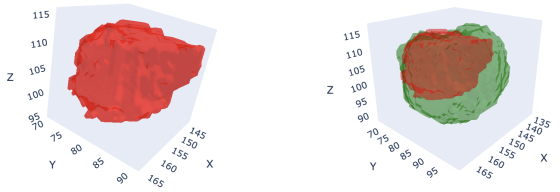


Figure 2: 3D volumetric representation of the valve artifact. (Left) Valve artifact estimated using geometric approach. (Right) Overlay of the prediction (red) on the ground truth mask (green).

The geometric subtraction is inherently constrained by the brain boundary, causing any extracranial portion of the signal void to be omitted by the convex hull subtraction, systematically underestimating the full artifact extent. To recover this, the partial geometric mask is used as a spatial prior for a 3D Residual U-Net (Alom et al., 2019), acting as a high-confidence spatial seed that reduces the network's search space to the intracranial region.

The refinement network is a dual-channel 3D Residual U-Net that fuses MRI intensities with the geometric prior across four hierarchical levels with feature depths of (32, 64, 128, 256) and strided downsampling of (2, 2, 2). Each block contains two residual units to ensure stable gradient flow, allowing the network to learn the residual difference between the conservative prior and the ground truth rather than segmenting from scratch. Training uses

a hybrid Dice and Cross-Entropy loss to address class imbalance, yielding a final segmentation that captures the artifact across both intracranial and extracranial regions.

4.3 Catheter Segmentation

Following artifact segmentation, the pipeline targets the shunt catheter, a thin, curvilinear structure representing only 0.02% of the total volume. Ground-truth masks contain three labels; background, visible catheter and occluded catheter, of which only the visible portion was treated as foreground during training, ensuring the model focuses on detectable structures.

The backbone is a 3D U-Net with four resolution levels, base channel width of 16, and $3\times3\times3$ convolutions with Instance Normalisation at each level. Training used $64\times64\times64$ patches sampled from the full $192\times224\times192$ volume, with 50% of patches centred on a catheter voxel to counteract extreme class imbalance and maintain a consistent gradient signal.

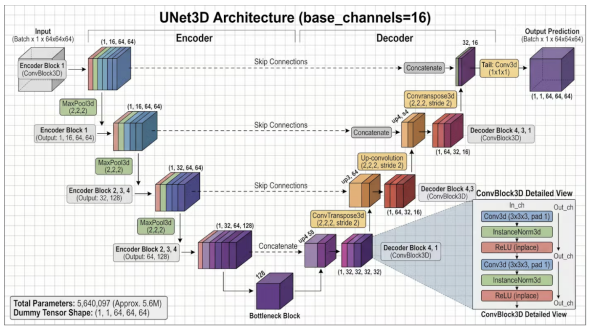


Figure 3: Detailed architecture of the 3D U-Net employed for catheter segmentation.

Optimization was performed over 30 epochs using the Adam optimizer and a ReduceLROnPlateau scheduler. To further force the model to prioritize the thin, curvilinear catheter structure, we utilized a hybrid Loss function combining Dice Loss with a weighted Binary Cross-Entropy (BCE). The BCE term incorporated a positive class weight of 50, heavily penalizing false negatives.

During inference, a sliding-window strategy was applied with a stride of 32 voxels to generate full-volume predictions. The resulting overlapping probability maps were averaged and thresholded at 0.7, a value determined via validation-set performance to achieve the optimal balance between precision and recall for the visible catheter.

4.4 Catheter Length Estimation

Catheter length estimation is complicated by a susceptibility artifact at the metallic tip that obscures part of the catheter in MRI. Total length is therefore decomposed as:

$$\hat{L}_{\text{total}} = L_{\text{visible}} + L_{\text{hidden}} \tag{3}$$

Estimation of the visible length is performed using a sequence of steps. **1**, the catheter mask is downsampled to one point per 2 mm to remove any staircase noise. **2**, a proximity graph is constructed by connecting nearby voxels within 2 mm. **3**, the minimum spanning tree (MST) of the graph is computed and the longest path through it is found via double BFS. This path defines the catheter spine (Figure **??**). **4**, a cubic B-spline is fitted to the spine and arc length is integrated at high resolution, reducing staircase overestimation by $\sim$15 % versus direct point summation. **5**, Spine points falling inside the artifact mask are trimmed to prevent overlap with L_{hidden}.

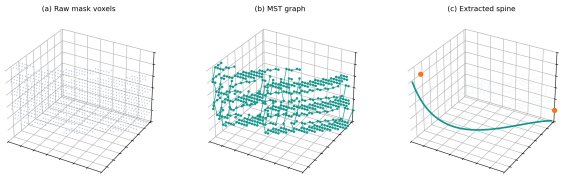


Figure 4: 3D visualisation of the MST graph and extracted catheter spine: (a) raw mask voxels, (b) MST graph, (c) final spine after double-BFS.

Estimation of the hidden length using geometric approaches proved unreliable due to artifact shape variability. Instead, we train a **Gradient Boosting Regressor** to predict the residual $L_{\text{hidden}} = \max(0,\, L_{\text{gt}} - L_{\text{visible}})$ from the features listed in Table 1, with hyperparameters: 1000 estimators, maximum depth 5, and learning rate 0.01. The model is validated via 5-fold cross-validation on L_{hidden} directly.

5 Results

5.1 Shunt Valve Artifact Segmentation

Quantitative metrics demonstrate a significant improvement in recovering the full volumetric extent of the hardware signal void when comparing the 3D Residual U-Net refinement against the initial geometric estimation (Table 2).

Table 1: Features used for hidden length regression.

Feature	Description
L_{visible}	Visible arc length (mm)
$\bar{\theta}$	Mean inter-segment angle (deg)
$\theta_{\max}$	Max inter-segment angle (deg)
V_{art}	Artifact mask volume (mm^3)
e_{art}	Artifact extent along tip direction (mm)
d_{tip}	Tip to nearest artifact voxel (mm)
N_{art}	Number of artifact voxels
d_{end}	Spine endpoint separation (mm)
N_{spine}	Spine points after trimming

Table 2: Performance comparison for valve artifact segmentation.

Method	Mean Dice	Max Dice	Mean FP	Mean FN
Geometric Estimation	0.373	0.731	**381.0**	5,372.0
Residual U-Net	**0.611**	**0.911**	1,861.7	**3,041.2**

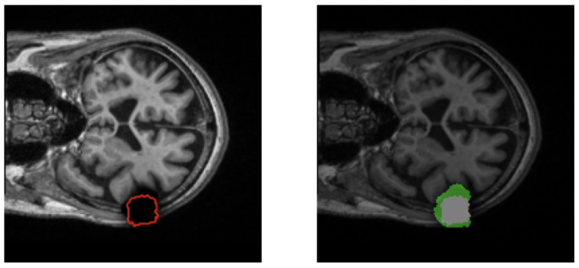


Figure 5: Qualitative assessment of artifact segmentation. (Left) Valve artifact prediction using U-Net model. (Right) Overlay of the prediction (gray) on the ground truth mask (green).

5.2 Catheter Segmentation

Quantitative evaluation on the validation set ($n = 40$) yielded a **Mean Dice Similarity Coefficient (DSC) of** 0.8165 ± 0.0602. The high sensitivity of the model to the thin, curvilinear structure is attributed to the positive sampling strategy and the heavily weighted cross-entropy loss, which prevented the network from defaulting to background predictions.

5.3 Catheter Length Estimation

Across all subjects, the method achieved a mean absolute error of 1.73 mm, a median absolute error of 0.03 mm, and a standard deviation of 4.99 mm. The low median indicates that the pipeline produces accurate estimates for the majority of cases, with the discrepancy between mean and median suggesting the presence of a small number of outlier subjects with larger errors.

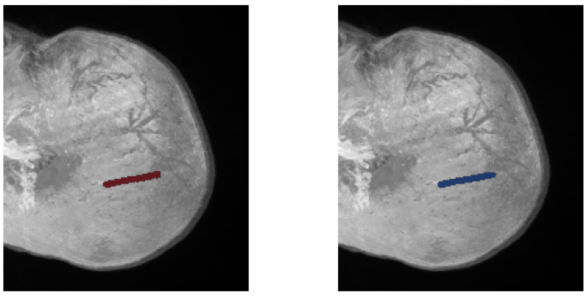


Figure 6: Qualitative assessment of catheter segmentation. The image shows the model prediction (left) and the ground-truth (right) across a representative MRI slice. The model effectively maintains topological continuity of the thin tube-like structure throughout the intracranial path.

6 Discussion and Conclusion

Despite the intent to anchor the model and reduce the search space, our geometric prior likely introduced a restrictive bottleneck, biasing predictions toward the intracranial region and constraining the network from recovering distal voxels near the artifact boundary. Teams training directly on raw MRI intensities benefited from broader global context, and one team notably proposed a W-Net architecture (Li, 2025) — a cascaded design we could leverage to refine first-stage predictions without any geometric prior, recovering missed voxels through learned intensity cues alone. Future iterations should also explore soft distance transform maps as a more flexible alternative to binary priors.

Our catheter segmentation achieved a second-rank Dice score of 0.8165 ± 0.0602, with validation plateauing in late training. Several teams reported gains by incorporating the artifact mask as a second input channel, localising precisely where the catheter terminates — the region where our model struggles most — and we hypothesise this two-channel configuration could break our current plateau alongside clDice loss and deep supervision to improve topological continuity.

The elevated length estimation error (MAE = 1.73 mm, std = 4.99 mm) reflects a cascade of segmentation errors; when the artifact mask is underestimated the regressor loses its spatial anchor, producing unstable predictions. Improving artifact segmentation robustness is therefore the primary prerequisite for stabilising downstream length estimation performance.

7 Contribution

- **Maaz Shamim:** End-to-end pipeline for Artifact Segmentation and Catheter Length Estimation including preprocessing, training and evaluation
- **Jadin Singh:** End-to-end pipeline for Catheter Segmentation including preprocessing, training and evaluation

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